2024, Mar. 2024 Release LOC:Acute Adult
Infection: Skin

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview

Instruction:
This subset is for patients with skin infections including cutaneous abscesses, herpes zoster and herpes zoster ophthalmicus (HZO), wound or surgical site infection, and diabetic foot ulcers. For patients with cellulitis refer to Infection: Cellulitis subset, patients with osteomyelitis refer to Infection: Musculoskeletal subset, patients with necrotizing fasciitis refer to General Surgical subset, or patients with actual or suspected sepsis refer to Infection: Sepsis subset.
Introduction:
Abscess, cutaneous: Cutaneous abscesses are purulent fluid collections that develop within the layers of skin tissue. Symptoms typically include pain and a fluctuant, red-purplish, raised area on the skin and may overlay an area of induration. Most abscesses are treated with incision and drainage alone. Anti-infective therapy is reserved for patients with multiple or difficult-to-drain abscesses, those with systemic signs of infection, and patients who are immunocompromised or older, or require additional incision and drainage (Schmitz and Gottlieb, Ann Emerg Med 2021, 78: 44-8; Stevens et al., Clin Infect Dis 2014, 59: e10-52).
Diabetic foot ulcer: A diabetic foot ulcer is a wound or lesion that develops due to peripheral neuropathy, neuro-osteoarthropathic deformities (e.g., Charcot disease), vascular insufficiency, patient disability, nonadherence to a foot care program, or lack of patient education or comprehension. Careful assessment of the wound is recommended to determine the severity and management of the infection. Patients with severe infections (e.g., gangrene, necrosis) or chronic limb-threatening ischemia should be hospitalized. Management includes intravenous anti-infective treatment, debridement, assessment of vascular supply, glucose control, and developing a wound care plan (Bolton, Wounds 2022, 34: 175-7; Lipsky et al., Diabetes Metab Res Rev 2020, 36 Suppl 1: e3280; National Institute for Clinical Excellence (NICE), Diabetic foot problems: prevention and management. 2015 (updated 2019)).
Herpes zoster/HZO: Herpes zoster, or commonly known as shingles, is caused by the reactivation of the varicella virus from a latent infection and presents as a painful unilateral, dermatomal, red, or maculopapular rash (often with vesicles). HZO manifests as conjunctivitis, uveitis, episcleritis, keratitis, or retinitis. Herpes zoster is commonly managed in the outpatient setting. However, hospitalization may be warranted if the rash crosses the midline of the body, is widespread (disseminated), if the patient is immunocompromised, or the herpatic rash is co-infected with a superimposed bacterial skin infection. HZO is an ophthalmologic emergency and requires hospitalization for specialty consultation and management as this condition can cause blindness. Diagnosis can be made based on clinical presentation, and treatment for patients who require hospitalization includes intravenous anti-infectives and pain management (Patil et al., Viruses 2022, 14:; Ahronowitz and Fox, J Am Acad Dermatol 2018; 78(1): 223-230 e223).

Wound or surgical site infection: Patients who present with classic indicators of a wound infection may have erythema, warmth, swelling, purulent discharge, unexpected delays in wound healing, new or worsening pain, or a malodorous wound odor. A spreading infection (e.g., an extension of erythema and induration around the wound, lymphangitis), dehiscence, a wound requiring debridement, or suspected systemic involvement may require hospitalization for evaluation and treatment. Management includes intravenous anti-infective treatment based on microbiology results, debridement, and establishing a wound care regimen (Swanson et al., J Wound Care 2022, 31: S10-s21).
Surgical site infections (SSIs) are deep tissue infections that occur at an incision site less than 30 days old (if overlying a prosthesis, then less than one-year-old). Treatment for this condition is suture removal and incision, and drainage. Anti-infectives may not be required; however, patients exhibiting systemic signs and symptoms should receive intravenous anti-infectives. Hospitalization is recommended when there is the presence of fever, elevated heart rate or white blood cell (WBC) count, or a large area of erythema and induration around the wound (Pinchera et al., Antibiotics (Basel) 2022, 11: epub; Stevens et al., Clin Infect Dis 2014, 59: e10-52).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook.* The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The majority of the content in this subset is based on a limited number of key sources:

- Abscess, cutaneous (Schmitz and Gottlieb, Ann Emerg Med 2021, 78: 44-8; Stevens et al., Clin Infect Dis 2014, 59: e10-52)
- Herpes zoster (Patil et al., Viruses 2022, 14:; Ahronowitz and Fox, J Am Acad Dermatol 2018; 78(1): 223-230 e223)
- Diabetic foot ulcer (Bolton, Wounds 2022, 34: 175-7; Lipsky et al., Diabetes Metab Res Rev 2020, 36 Suppl 1: e3280; National Institute for Clinical Excellence (NICE), Diabetic foot problems: prevention and management. 2015 (updated 2019))
- Surgical site infection (Pinchera et al., Antibiotics (Basel) 2022, 11: epub; Stevens et al., Clin Infect Dis 2014, 59: e10-52)
- Wound Infection (Swanson et al., J Wound Care 2022, 31: S10-s21)

(Excludes PO medications unless noted)

Select Day, One:

- **Initial review, One:** (1, 2)
  - (From arrival in ED through decision to admit)
  - (Symptom or finding within 24h)
  - **OBSERVATION, ≥ One:** (3)
    - Abscess, cutaneous and, ≥ **One:** (4)
      - Systemic infection suspected, **Both:**
        - SIRS, ≥ **Two:** (5)
          - Temperature ≥ 100.4°F(38.0°C) or < 97.0°F(36.1°C) (6)
          - WBC, ≥ **One:**
            - > 12,000/cu.mm(12x10^9/L) or < 4,000/cu.mm(4x10^9/L)
            - Bands > 10%(0.10)
          - Heart rate > 90/min, sustained (7)
          - Respiratory rate > 20/min, sustained (7)
        - Lactic acid ≤ ULN
      - Failed outpatient treatment, **Both:** (8)
        - Incision and drainage performed prior to hospitalization and no evidence of clinical improvement
        - Failed outpatient anti-infective (includes PO)
      - Immunocompromised (9, 10)
    - Herpes zoster and, **Both:** (11)
      - Failed outpatient anti-infective treatment (includes PO), ≥ **One:** (8)
        - Continued deterioration despite ≥ 24h treatment (includes PO) (12)
        - Non-compliant with outpatient anti-infective (includes PO)
        - Unable to tolerate or absorb oral anti-infective
        - Unresponsive to outpatient anti-infective (includes PO), ≥ **One:** (13)
          - ≥ 3d
          - ≥ 5 doses
      - Anti-infective
    - Wound or surgical site infection and, ≥ **One:** (14)
      - Dehiscence
      - Failed outpatient anti-infective treatment (includes PO), ≥ **One:** (8)
        - Continued deterioration despite ≥ 24h treatment (includes PO) (12)
        - Non-compliant with outpatient anti-infective (includes PO)
        - Unable to tolerate or absorb oral anti-infective
        - Unresponsive to outpatient anti-infective (includes PO), ≥ **One:** (13)
          - ≥ 3d
          - ≥ 5 doses
      - Purulent incisional drainage and wound culture pending
      - Requiring re-exploration or debridement
      - Systemic infection suspected, **Both:**
        - SIRS, ≥ **Two:** (5)
          - Temperature ≥ 100.4°F(38.0°C) or < 97.0°F(36.1°C) (6)
          - WBC, ≥ **One:**
            - > 12,000/cu.mm(12x10^9/L) or < 4,000/cu.mm(4x10^9/L)
            - Bands > 10%(0.10)
          - Heart rate > 90/min, sustained (7)
          - Respiratory rate > 20/min, sustained (7)
        - Lactic acid ≤ ULN
      - Wound culture, positive (includes outpatient specimen)
  - **ACUTE, ≥ One:** (15)
    - Diabetic foot ulcer and, **Both:** (16)
      - Finding, ≥ **One:**

- AKI [17]
- Chronic limb-threatening ischemia (CLTI) or impaired vascularity suspected
- Deep tissue abscess or infection
- Failed outpatient anti-infective (includes PO), ≥ **One:** [18]
    - Continued deterioration despite ≥ 24h treatment (includes PO) [12]
    - Unable to tolerate or absorb oral anti-infective
    - Unresponsive to outpatient anti-infective (includes PO), ≥ **One:** [13]
        - ≥ 3d
        - ≥ 5 doses
- Gangrene
- Immunocompromised [9]
- Involvement of muscle, tendon, joint, or bone
- Lymphedema
- Necrosis
- Presence of foreign body

- Anti-infective

- Herpes zoster and, **Both:** [19]
    - Finding, ≥ **One:**
        - Disseminated
        - Immunocompromised [9]
        - Ophthalmicus [20]
    - Anti-infective
- Wound or surgical site infection and, ≥ **One:** [21, 22]
    - Temperature ≥ 101.3°F(38.5°C) [23]
    - Erythema and induration > 5 cm from wound edge
    - Heart rate > 110/min, sustained [7]
    - Located over a prosthesis or implanted device [24]
    - WBC, ≥ **One:**
        - > 12,000/cu.mm(12x10^9/L) or < 4,000/cu.mm(4x10^9/L)
        - Bands > 10%(0.10)

- **Episode Day 1, One:** [1]
  (Symptom or finding within 24h)
    - **OBSERVATION, ≥ One:** [3]
        - Cellulitis (**use Infection: Cellulitis subset**)
        - Abscess, cutaneous and, **Both:** [4]
            - Finding, ≥ **One:**
                - Systemic infection suspected, **Both:**
                    - SIRS, ≥ **Two:** [5]
                        - Temperature ≥ 100.4°F(38.0°C) or < 97.0°F(36.1°C) [6]
                        - WBC, ≥ **One:**
                            - > 12,000/cu.mm(12x10^9/L) or < 4,000/cu.mm(4x10^9/L)
                            - Bands > 10%(0.10)
                        - Heart rate > 90/min, sustained [7]
                        - Respiratory rate > 20/min, sustained [7]
                    - Lactic acid ≤ ULN
                - Failed outpatient treatment, **Both:** [8]
                    - Incision and drainage performed prior to hospitalization and no evidence of clinical improvement
                    - Failed outpatient anti-infective (includes PO)
                - Immunocompromised [9, 10]
            - Intervention, ≥ **One:**
                - Anti-infective [25]
                - Incision and drainage [26]
        - Herpes zoster and, **Both:** [11]

- Failed outpatient anti-infective treatment (includes PO), ≥ **One:** (8)
  - Continued deterioration despite ≥ 24h treatment (includes PO) (12)
  - Non-compliant with outpatient anti-infective (includes PO)
  - Unable to tolerate or absorb oral anti-infective
  - Unresponsive to outpatient anti-infective (includes PO), ≥ **One:** (13)
    - ≥ 3d
    - ≥ 5 doses
- Anti-infective

- Wound or surgical site infection and, **Both:** (14)
  - Finding, ≥ **One:**
    - Dehiscence
    - Failed outpatient anti-infective treatment (includes PO), ≥ **One:** (8)
      - Continued deterioration despite ≥ 24h treatment (includes PO) (12)
      - Non-compliant with outpatient anti-infective (includes PO)
      - Unable to tolerate or absorb oral anti-infective
      - Unresponsive to outpatient anti-infective (includes PO), ≥ **One:** (13)
        - ≥ 3d
        - ≥ 5 doses
    - Purulent incisional drainage and wound culture pending
    - Requiring re-exploration or debridement
    - Systemic infection suspected, **Both:**
      - SIRS, ≥ **Two:** (5)
        - Temperature ≥ 100.4°F(38.0°C) or < 97.0°F(36.1°C) (6)
        - WBC, ≥ **One:**
          - > 12,000/cu.mm($12x10^9$/L) or < 4,000/cu.mm($4x10^9$/L)
          - Bands > 10%(0.10)
        - Heart rate > 90/min, sustained (7)
        - Respiratory rate > 20/min, sustained (7)
      - Lactic acid ≤ ULN
    - Wound culture, positive (includes outpatient specimen)
  - Intervention, ≥ **One:**
    - Anti-infective
    - Incision and drainage or suture removal

- **ACUTE, ≥ One:** (15)
  - Cellulitis (**use Infection: Cellulitis subset**)
  - Diabetic foot ulcer and, **All:** (16)
    - Finding, ≥ **One:**
      - AKI (17)
      - Chronic limb-threatening ischemia (CLTI) or impaired vascularity suspected
      - Deep tissue abscess or infection
      - Failed outpatient anti-infective (includes PO), ≥ **One:** (18)
        - Continued deterioration despite ≥ 24h treatment (includes PO) (12)
        - Unable to tolerate or absorb oral anti-infective
        - Unresponsive to outpatient anti-infective (includes PO), ≥ **One:** (13)
          - ≥ 3d
          - ≥ 5 doses
      - Gangrene
      - Immunocompromised (9)
      - Involvement of muscle, tendon, joint, or bone
      - Lymphedema
      - Necrosis
      - Presence of foreign body
    - Assessment of vascular supply, ≥ **One:**

- Vascular imaging [27]
- Noninvasive evaluation [28, 29]
- Performed within the last 3 mos
- Imaging to rule out osteomyelitis [30, 31]
- Anti-infective
- Surgical evaluation for debridement
- Herpes zoster and, **Both:** [19]
  - Finding, ≥ **One:**
    - Disseminated
    - Immunocompromised [9]
    - Ophthalmicus [20]
  - Anti-infective
- Wound or surgical site infection and, **Both:** [21, 22]
  - Finding, ≥ **One:**
    - Temperature ≥ 101.3°F(38.5°C) [23]
    - Erythema and induration > 5 cm from wound edge
    - Heart rate > 110/min, sustained [7]
    - Located over a prosthesis or implanted device [24]
    - WBC, ≥ **One:**
      - > 12,000/cu.mm($12x10^9$/L) or < 4,000/cu.mm($4x10^9$/L)
      - Bands > 10%(0.10)
  - Intervention, ≥ **One:**
    - Anti-infective
    - Incision and drainage or suture removal
- **Episode Day 2, One:** [32]
  - **OBSERVATION, ≥ One:**
    - Abscess, cutaneous and, **One:**
      - **Responder**, discharge expected today if clinically stable last 12h, **All:** [33]
        - Afebrile
        - Anti-infective, **One:**
          - Arranged
          - Completed
          - Not indicated
        - Laboratory values within acceptable limits [34]
        - Vital signs within acceptable limits [34]
        - Wound care, ≥ **One:**
          - Wound care regimen established
          - Wound healing or manageable
      - **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
        - Intervention, ≥ **One:** [35]
          - IV anti-infective
          - Transition to PO anti-infective ≤ 24h
          - Wound care
    - Herpes zoster and, **One:**
      - **Responder**, discharge expected today if clinically stable last 12h, **One:**
        - Anti-infective regimen established and tolerated
      - **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
        - Intervention, ≥ **One:** [35]
          - IV anti-infective
          - Transition to PO anti-infective ≤ 24h
    - Wound or surgical site infection and, **One:**
      - **Responder**, discharge expected today if clinically stable last 12h, **All:**
        - Afebrile

- Anti-infective, **One:**
  - Arranged
  - Completed
  - Not indicated
- Laboratory values within acceptable limits [34]
- Vital signs within acceptable limits [34]
- Wound care, ≥ **One:**
  - Wound care regimen established
  - Wound healing or manageable

Partial responder, not clinically stable for discharge and requires continued stay, ≥ **One:**
- Anti-infective
- Transition to PO anti-infective ≤ 24h
- Wound care

ACUTE, One:
- **Early responder**, discharge expected today if clinically stable last 12h, **All:** [33]
  - Afebrile
  - Anti-infective regimen established and tolerated
  - Herpes zoster resolving
  - No complication or active comorbidity relevant to this episode of care [36]
- **Partial responder**, not clinically stable for discharge and requires continued stay, **One:**
  - Osteomyelitis (**use Infection: Musculoskeletal subset**)
  - Infection unresolved and, ≥ **One:**
    - Diabetic foot ulcer and anti-infective ≤ 8 days since initiation
    - Herpes zoster and anti-infective ≤ 5 days since initiation
    - Wound or surgical site infection and, ≥ **One:**
      - Anti-infective ≤ 7 days since initiation
      - Wound care

Episode Day 3, One: [37]
- **OBSERVATION,** ≥ **One:**
  - Abscess, cutaneous and, **One:**
    - **Responder,** discharge expected today if clinically stable last 12h, **All:** [33]
      - Afebrile
      - Anti-infective, **One:**
        - Arranged
        - Completed
        - Not indicated
      - Laboratory values within acceptable limits [34]
      - Vital signs within acceptable limits [34]
      - Wound care, ≥ **One:**
        - Wound care regimen established
        - Wound healing or manageable
    - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
      - For cutaneous abscess use Episode Day 3 at a higher level of care
      - For a condition other than cutaneous abscess use appropriate subset (Episode Day 1) based on clinical findings
      - Refer for secondary review
  - Herpes zoster and, **One:**
    - **Responder**, discharge expected today if clinically stable last 12h, **One:**
      - Anti-infective regimen established and tolerated
    - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
      - For herpes zoster use Episode Day 3 at a higher level of care

- For a condition other than herpes zoster use appropriate subset (Episode Day 1) based on clinical findings
- Refer for secondary review

- ⊖ Wound or surgical site infection and, **One:** (38)
  - ⊖ **Responder**, discharge expected today if clinically stable last 12h, **All:**
    - Afebrile
    - ⊖ Anti-infective, **One:**
      - Arranged
      - Completed
      - Not indicated
    - Laboratory values within acceptable limits (34)
    - Vital signs within acceptable limits (34)
    - ⊖ Wound care, ≥ **One:**
      - Wound care regimen established
      - Wound healing or manageable
  - ⊖ **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
    - For wound or surgical site infection use Episode Day 3 at a higher level of care
    - For a condition other than wound or surgical site infection use appropriate subset (Episode Day 1) based on clinical findings
    - Refer for secondary review

⊖ **ACUTE, One:**

- ⊖ **Responder,** discharge expected today if clinically stable last 12h, **All:** (33)
  - Afebrile
  - Anti-infective regimen established and tolerated
  - Herpes zoster resolving and no new lesions
  - ⊖ Complication or comorbidity, **One:**
    - No complication or active comorbidity relevant to this episode of care (36)
    - ⊖ Complication or comorbidity and clinically stable for discharge, ≥ **One:**
      - Adverse drug reaction resolving
      - Blood glucose within acceptable limits (34)
      - *C. difficile* infection and decrease in frequency and severity of diarrhea
      - Pain controlled or manageable (39)
- ⊖ **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
  - Abscess (cutaneous) worsening or unimproved and anti-infective ≤ 5d since initiation
  - ⊖ Diabetic foot ulcer and, **Both:**
    - ⊖ Finding, ≥ **One:**
      - Worsening or unimproved
      - Gangrene, unresolved
    - ⊖ Intervention, ≥ **One:**
      - Anti-infective ≤ 8 days since initiation
      - Hyperbaric oxygen (40)
  - Herpes zoster worsening or unimproved and anti-infective ≤ 5 days since initiation
  - ⊖ Wound or surgical site infection, worsening or unimproved, ≥ **One:**
    - Wound care
    - Anti-infective ≤ 7 days since initiation
  - Adverse anti-infective reaction requiring change in medication ≤ 24h (41)
  - ⊖ Blood glucose ≤ 24h, **All:** (42)
    - ⊖ Finding, ≥ **One:**
      - > 250 mg/dL(13.9 mmol/L)
      - < 70 mg/dL(3.9 mmol/L)
    - Blood glucose monitoring
    - Insulin adjustment or dose held (42, 43)

- *C. difficile* infection, actual or suspected and, **Both:** [44]
  - Anti-infective (includes PO) ≤ 5d [45, 46]
  - Inadequate oral intake [47]
- Complex wound care ≤ 5d and, ≥ **One:** [38]
  - Wound care ≥ 3x/24h and, ≥ **One:**
    - Requiring parenteral analgesic
    - ≥ 15 min in duration
  - Wound management teaching ≤ 24h [48]
- Pain uncontrolled and, **Both:**
  - Pain assessment [49]
  - Requiring change in medication, dose, or frequency [50, 51]

- **Episode Day 4-7, One:**
  - **ACUTE, One:**
    - **Responder,** discharge expected today if clinically stable last 12h, **All:** [33]
      - Afebrile
      - Anti-infective regimen established and tolerated
      - Infection and, ≥ **One:**
        - Abscess, resolving and wound manageable
        - Diabetic foot ulcer, resolving and wound manageable
        - Gangrene resolving and wound manageable
        - Herpes zoster, resolving and no new lesions
        - Wound or surgical site infection, resolving and wound manageable
      - Complication or comorbidity, **One:**
        - No complication or active comorbidity relevant to this episode of care [36]
        - Complication or comorbidity and clinically stable for discharge, ≥ **One:**
          - Adverse drug reaction resolving
          - Blood glucose within acceptable limits [34]
          - *C. difficile* infection and decrease in frequency and severity of diarrhea
          - Pain controlled or manageable [39]
    - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
      - Abscess (cutaneous) unresolved and anti-infective ≤ 5d since initiation
      - Diabetic foot ulcer and, **Both:**
        - Finding, ≥ **One:**
          - Worsening or unimproved
          - Gangrene, unresolved
        - Intervention, ≥ **One:**
          - Anti-infective ≤ 8 days since initiation
          - Hyperbaric oxygen [40]
      - Herpes zoster worsening or unimproved and anti-infective ≤ 5 days since initiation
      - Wound or surgical site infection and, **Both:**
        - Worsening or unimproved
        - Anti-infective ≤ 7 days since initiation
      - Adverse anti-infective reaction requiring change in medication ≤ 24h [41]
      - Blood glucose ≤ 24h, **All:** [42]
        - Finding, ≥ **One:**
          - > 250 mg/dL(13.9 mmol/L)
          - < 70 mg/dL(3.9 mmol/L)
        - Blood glucose monitoring
        - Insulin adjustment or dose held [42, 43]
      - *C. difficile* infection, actual or suspected and, **Both:** [44]
        - Anti-infective (includes PO) ≤ 5d [45, 46]
        - Inadequate oral intake [47]
      - Complex wound care ≤ 5d and, ≥ **One:** [38]

- Wound care ≥ 3x/24h and, ≥ **One:**
  - Requiring parenteral analgesic
  - ≥ 15 min in duration
- Wound management teaching ≤ 24h [48]
- Pain uncontrolled and, **Both:**
  - Pain assessment [49]
  - Requiring change in medication, dose, or frequency [50, 51]

- **Episode Day 8, One:**
  - **ACUTE, One:**
    - **Responder**, discharge expected today if clinically stable last 12h, **All:** [33]
      - Afebrile
      - Anti-infective regimen established and tolerated
      - Infection and, ≥ **One:**
        - Abscess, resolving and wound manageable
        - Diabetic foot ulcer, resolving and wound manageable
        - Gangrene resolving and wound manageable
        - Herpes zoster, resolving and no new lesions
        - Wound or surgical site infection, resolving and wound manageable
      - Complication or comorbidity, **One:**
        - No complication or active comorbidity relevant to this episode of care [36]
        - Complication or comorbidity and clinically stable for discharge, ≥ **One:**
          - Adverse drug reaction resolving
          - Blood glucose within acceptable limits [34]
          - *C. difficile* infection and decrease in frequency and severity of diarrhea
          - Pain controlled or manageable [39]
    - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
      - **Use Extended Stay subset**
      - Refer for secondary review

Discharge, Level of Care, One: [52]

- **HOME, Both:**
  - Level of care appropriateness, **Both:**
    - Home environment safe and accessible [53]
    - Patient or caregiver demonstrates ability to manage care
  - Discharge planning, **All:**
    - Provide comprehensive written discharge and teaching instructions [54]
    - Medication reconciliation [55]
    - Follow-up care planned [56]
    - Identify and address transportation needs
- **HOME CARE, Both:**
  - Level of care appropriateness, **All:**
    - Home environment safe and accessible [53]
    - Patient or caregiver willing and able to learn care
    - Outpatient management contraindicated or unavailable [57, 58]
    - Skilled services, ≥ **One:** [59]
      - Skilled nursing [60]
      - Skilled therapy, PT, OT, or ST
      - Behavioral health [61]
  - Discharge Planning, **All:**
    - Provide comprehensive written discharge and teaching instructions [54]
    - Medication reconciliation [55]
    - Follow-up care planned [56]

- Identify and address transportation needs

- **BEHAVIORAL HEALTH, Both:**
  - Level of care appropriateness, **One:**
    - Support systems available [62]
  - Skilled treatment, ≥ **One:**
    - Clinical assessment [63]
    - Individual, group, or family counseling
    - Psychiatric, substance, or medication evaluation [64]
- **SKILLED MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
    - Treatment precluded in a lower level
    - Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 1-2x/24h [60]
      - Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [65, 66]
        - Functional impairments requiring at least supervision [67]
        - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [55]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **SUBACUTE MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
    - Treatment precluded in a lower level
    - Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 4h/24h [60]
      - Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [65, 66]
        - ≥ 2 functional impairments requiring at least minimum assistance [67]
        - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [55]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **SUBACUTE REHAB, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Treatment precluded in a lower level
    - Services, **All:**
      - Goal directed therapy with rehabilitation potential [65, 66]
      - ≥ 2 functional impairments requiring at least minimum assistance [67]
      - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [55]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **ACUTE REHAB, Both:**
  - Level of care appropriateness, **Both:**

- Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
- ⊖ Services, **All:**
  - Goal directed therapy with rehabilitation potential [65, 66]
  - ≥ 2 functional impairments requiring at least minimum assistance [67]
  - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines [68]

⊖ Complete prior to facility transfer, **All:**
- Medication reconciliation [55]
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

⊖ **LONG TERM ACUTE CARE, Both:**

⊖ Level of care appropriateness, **Both:**
- Treatment precluded in a lower level
- ⊖ In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One:**
  - Primary condition or illness and treatment of active comorbid condition(s) [69]
  - Ventilator dependent ≥ 6h/d and weaning planned [70, 71]
  - Acute and comorbid conditions requiring prolonged hospitalization [72]

⊖ Complete prior to facility transfer, **All:**
- Medication reconciliation [55]
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
073 CRANIAL AND PERIPHERAL NERVE DISORDERS WITH MCC	**n/a**	**3.9**	**CMS GMLOS**
074 CRANIAL AND PERIPHERAL NERVE DISORDERS WITHOUT MCC	**n/a**	**2.9**	**CMS GMLOS**
121 ACUTE MAJOR EYE INFECTIONS WITH CC/MCC	**n/a**	**4.4**	**CMS GMLOS**
122 ACUTE MAJOR EYE INFECTIONS WITHOUT CC/MCC	**n/a**	**3**	**CMS GMLOS**
124 OTHER DISORDERS OF THE EYE WITH MCC OR THROMBOLYTIC AGENT	**n/a**	**3.4**	**CMS GMLOS**
125 OTHER DISORDERS OF THE EYE WITHOUT MCC	**n/a**	**2.3**	**CMS GMLOS**
154 OTHER EAR, NOSE, MOUTH AND THROAT DIAGNOSES WITH MCC	**n/a**	**4.2**	**CMS GMLOS**
155 OTHER EAR, NOSE, MOUTH AND THROAT DIAGNOSES WITH CC	**n/a**	**2.9**	**CMS GMLOS**
156 OTHER EAR, NOSE, MOUTH AND THROAT DIAGNOSES WITHOUT CC/MCC	**n/a**	**2.1**	**CMS GMLOS**
157 DENTAL AND ORAL DISEASES WITH MCC	**n/a**	**4.7**	**CMS GMLOS**
158 DENTAL AND ORAL DISEASES WITH CC	**n/a**	**2.9**	**CMS GMLOS**

Condition or Procedure	% Pd Obs	LOS (days)	Type
159 DENTAL AND ORAL DISEASES WITHOUT CC/MCC	n/a	2	CMS GMLOS
299 PERIPHERAL VASCULAR DISORDERS WITH MCC	n/a	4	CMS GMLOS
300 PERIPHERAL VASCULAR DISORDERS WITH CC	n/a	3.1	CMS GMLOS
301 PERIPHERAL VASCULAR DISORDERS WITHOUT CC/MCC	n/a	2.1	CMS GMLOS
393 OTHER DIGESTIVE SYSTEM DIAGNOSES WITH MCC	n/a	4.4	CMS GMLOS
394 OTHER DIGESTIVE SYSTEM DIAGNOSES WITH CC	n/a	3	CMS GMLOS
395 OTHER DIGESTIVE SYSTEM DIAGNOSES WITHOUT CC/MCC	n/a	2.1	CMS GMLOS
592 SKIN ULCERS WITH MCC	n/a	5.9	CMS GMLOS
593 SKIN ULCERS WITH CC	n/a	4.3	CMS GMLOS
594 SKIN ULCERS WITHOUT CC/MCC	n/a	3.1	CMS GMLOS
600 NON-MALIGNANT BREAST DISORDERS WITH CC/MCC	n/a	3.4	CMS GMLOS
601 NON-MALIGNANT BREAST DISORDERS WITHOUT CC/MCC	n/a	2.5	CMS GMLOS
606 MINOR SKIN DISORDERS WITH MCC	n/a	4.7	CMS GMLOS
607 MINOR SKIN DISORDERS WITHOUT MCC	n/a	3	CMS GMLOS
637 DIABETES WITH MCC	n/a	4.1	CMS GMLOS
638 DIABETES WITH CC	n/a	3	CMS GMLOS
639 DIABETES WITHOUT CC/MCC	n/a	2.1	CMS GMLOS
727 INFLAMMATION OF THE MALE REPRODUCTIVE SYSTEM WITH MCC	n/a	4.9	CMS GMLOS
728 INFLAMMATION OF THE MALE REPRODUCTIVE SYSTEM WITHOUT MCC	n/a	2.9	CMS GMLOS
729 OTHER MALE REPRODUCTIVE SYSTEM DIAGNOSES WITH CC/MCC	n/a	3	CMS GMLOS
730 OTHER MALE REPRODUCTIVE SYSTEM DIAGNOSES WITHOUT CC/MCC	n/a	1.9	CMS GMLOS
757 INFECTIONS, FEMALE REPRODUCTIVE SYSTEM WITH MCC	n/a	5	CMS GMLOS

Condition or Procedure	% Pd Obs	LOS (days)	Type
758 INFECTIONS, FEMALE REPRODUCTIVE SYSTEM WITH CC	n/a	3.7	CMS GMLOS
759 INFECTIONS, FEMALE REPRODUCTIVE SYSTEM WITHOUT CC/MCC	n/a	2.6	CMS GMLOS
862 POSTOPERATIVE AND POST-TRAUMATIC INFECTIONS WITH MCC	n/a	5.1	CMS GMLOS
863 POSTOPERATIVE AND POST-TRAUMATIC INFECTIONS WITHOUT MCC	n/a	3.5	CMS GMLOS
867 OTHER INFECTIOUS AND PARASITIC DISEASES DIAGNOSES WITH MCC	n/a	5.5	CMS GMLOS
868 OTHER INFECTIOUS AND PARASITIC DISEASES DIAGNOSES WITH CC	n/a	3.7	CMS GMLOS
869 OTHER INFECTIOUS AND PARASITIC DISEASES DIAGNOSES WITHOUT CC/MCC	n/a	2.5	CMS GMLOS
ABSCESS, CUTANEOUS	25%	4.1	InterQual
DIABETIC FOOT ULCER	23%	5.7	InterQual
HERPES ZOSTER	31%	3.8	InterQual
WOUND INFECTION	17%	5.7	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:

1:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.
For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.

- **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.
Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.
Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.
For further clarification, see the Acute Level of Care Review Process.

3:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

4:

Most abscesses are treated with incision and drainage alone. Anti-infective therapy is reserved for patients with multiple or difficult-to-drain abscesses, those with systemic signs of infection, and patients who are immunocompromised or older, or require additional incision and drainage (Schmitz and Gottlieb, Ann Emerg Med 2021, 78: 44-8; Stevens et al., Clin Infect Dis 2014, 59: e10-52).

5:

Systemic inflammatory response syndrome (SIRS) is an inflammatory process defined as 2 or more of the following clinical findings: fever, leukocytosis, tachypnea, and tachycardia.

6:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

7:

Vital signs are considered sustained findings when they are abnormal for 2 or more readings greater than 15 minutes apart or are lasting at least 15 minutes. This excludes an isolated reading, a transient abnormal measurement, or a finding that requires urgent treatment (e.g., severe hypoxemia, ventricular arrhythmia, hypotension). Application of criteria for a vital sign finding should be based on confirmatory measurements repeated at regular intervals. Events that are infrequent or vital sign abnormalities that have resolved with outpatient treatment are not considered to be sustained (e.g., resolution of hypertension following anti-hypertensive therapy, resolution of tachycardia following rehydration).
Note: The definition of sustained reflects the opinion of InterQual's® external peer reviewers. This criteria point is based upon current best practice and is the product of an iterative process involving multiple clinicians with diverse expertise in varied geographic settings.

8:

Instruction: This criteria point does not include failed treatment at the Observation level of care.
These criteria refer to the lack of response to treatment received through the medical practitioner's office, outpatient clinic, trial of home care services, or emergency department. These criteria can be applied when the patient fails to respond to more intensive treatment beyond the routine treatments or medications delivered for a chronic condition.

9:

Immunocompromised refers to the patient who has a decreased capacity to defend against invading organisms as a result of an impaired immune system due to several comorbidities. The principal manifestation of immunodeficiency is increased susceptibility to infection with unusual severity, frequency, and duration which differs from that of an individual who is immunocompetent who may be exposed to the same pathogen. Furthermore, individuals who are immunocompromised are increasingly prone to opportunistic infections. Immunodeficiency is classified as either primary (congenital), caused by inherited disorders of the immune system, or secondary (acquired), as a result of a disease process or as a result of treatment of other chronic diseases.

Primary causes of immunodeficiency account for more than 200 genetic disorders that impact immune system function. Some examples of primary immunodeficiency include congenital immunodeficiency diseases (e.g., X-linked agammaglobulinemia), combined immunodeficiency disease (e.g., DiGeorge syndrome, Wiskott syndrome), and chronic granulomatous disease.
Examples of secondary immune deficiency include poorly controlled diabetes mellitus defined as HbA1c 9%(0.09) (drawn on admission or within the past three months), end-stage liver disease, hematopoietic malignancies, patients who have had allogeneic hematopoietic stem cell transplantation (HSCT), and patients receiving immunosuppressive therapy (e.g., chemotherapy, radiation, anti-rejection medications, or prolonged high steroid use). Patients with severe, prolonged neutropenia (e.g., absolute neutrophil count less than 500/cu.mm(500x106/L), HSCT, human immunodeficiency virus (HIV) with CD4 count less than 200/cu.mm(200x106/L), a history of splenectomy, splenic dysfunction due to sickle cell disease, and those who have received intense chemotherapy (e.g., childhood acute myelogenous leukemia) are at greatest risk of infection (Centers for Disease Control and Prevention, National Diabetes Statistics Report. 2022; Bonilla et al., J Allergy Clin Immunol 2015, 136: 1186-205 e78; Lipska et al., Diabetes Care 2013, 36: 3535-42).

10:

Patients who are immunocompromised presenting with a cutaneous abscess are at high risk for treatment failure with incision and drainage alone. These patients commonly require intravenous anti-infectives to aid in resolution of the abscess (Stevens et al., Clin Infect Dis 2014, 59: e10-52).

11:

Herpes zoster is commonly managed in the outpatient setting. Treatment for patients who require hospitalization includes intravenous anti-infectives and pain management (Patil et al., Viruses 2022, 14:; Ahronowitz and Fox, J Am Acad Dermatol 2018; 78(1): 223-230 e223).

12:

Patients experiencing continued deterioration of their infection demonstrate symptoms which may include fever, pain, mental status change, hypotension, hypoxia, nausea, and/or vomiting.

13:

A patient should demonstrate objective signs of clinical improvement within 72 hours of initiation of treatment.

14:

Hospitalization may be necessary for patients who present with a wound or surgical site infection that has dehisced, has purulent drainage, requires re-exploration, having systemic involvement, or a positive wound culture. Management includes symptom stabilization, intravenous anti-infective treatment based on microbiology findings, suture removal or incision and drainage, and the development of a wound care regimen (Swanson et al., J Wound Care 2022, 31: S10-s21).

15:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

16:

Assessing the severity of infection in a patient with a diabetic foot ulcer becomes vital to risk stratify the patient who warrants hospitalization for diagnostic approaches and appropriate inpatient therapies. Diabetic foot infection (DFI) is the most frequent complication of diabetes requiring hospitalization and commonly leads to lower extremity amputation. According to the most recent guidelines by the Infectious Diseases Society of America (IDSA) and International Working Group on the Diabetic Foot (IWGDF), when certain risk factors or complicating features are present, this could signify or suggest a moderate or severe DFI requiring urgent hospitalization, especially in the presence of peripheral arterial disease (Senneville et al., Clin Infect Dis 2023: epub). These risk factors may include (but are not limited to) (Senneville et al., Clin Infect Dis 2023: epub):

- Acute kidney injury
- Deep tissue abscess
- Failed outpatient anti-infective therapy
- Gangrene or necrosis
- Presence of foreign body
- Immunocompromised state

Management for DFI includes intravenous anti-infective treatment(s), debridement if necessary, vascular assessment, optimal glucose control, and developing a wound care plan (Bolton, Wounds 2022, 34: 175-7; Lipsky et al., Diabetes Metab Res Rev 2020, 36 Suppl 1: e3280).

17:

Acute kidney injury (AKI) is a term used to describe the broad spectrum of kidney function impairment, including insufficiency, oliguria, and failure. The spectrum ranges from minor changes in renal function markers (e.g., increase in serum creatinine, decreased urine output) to failure requiring renal replacement therapy.
AKI is defined as a sudden decrease in renal function resulting from multiple etiologies, including intrinsic kidney disease, ischemia, nephrotoxicity, and extrarenal pathology.
AKI encompasses patients with chronic kidney disease who experience an acute deterioration. AKI is associated with significant morbidity and mortality in hospitalized patients (Ostermann et al., Kidney Int 2020, 98: 294-309; Guideline Updates, In: Acute kidney injury: prevention, detection and management. 2019; Kidney Disease: Improving Global Outcomes (KDIGO), Kidney International Supplements 2012, 2: ii.-138; Wang et al., Am J Nephrol 2012, 35: 349-55).

18:

These criteria refer to the lack of response to treatment received through the medical practitioner's office, outpatient clinic (e.g., urgent care), teleconsultation, trial of home care services, emergency department (prior to decision to admit), or hospital Observation setting (if appropriate for this condition). These criteria can be applied when the patient fails to respond to more intensive treatment beyond the routine treatments or medications delivered for a chronic condition.

19:

Herpes zoster is commonly managed in the outpatient setting. Hospitalization may be warranted if the rash crosses the midline of the body and is widespread (disseminated), on the face, or if the patient is immunocompromised (Patil et al., Viruses 2022, 14:; Ahronowitz and Fox, J Am Acad Dermatol 2018; 78(1): 223-230 e223).

20:

Herpes zoster ophthalmicus (HZO) is a variant of herpes zoster that manifests as a rash on the face, nose (e.g., Hutchinson sign), or around the eyes. Due to the risk of blindness, HZO is an ophthalmologic emergency that requires hospitalization for prompt ophthalmology consultation and the initiation of antiviral agents (Patil et al., Viruses 2022, 14:).

21:

Patients presenting with a surgical site infection (SSI) within 30 days of the original procedure may show signs and symptoms of erythema, fluctuance, warmth, and induration surrounding the surgical site. SSI also includes areas of infection overlying a surgical site where a prosthesis may have been implanted within the last year. If patients present with systemic involvement, they may require hospitalization for treatment, including incision and drainage, debridement, and anti-infectives (Pinchera et al., Antibiotics (Basel) 2022, 11: epub; Swanson et al., J Wound Care 2022, 31: S10-s21; Stevens et al., Clin Infect Dis 2014, 59: e10-52).

22:

Instruction: The infection must have occurred within 30 days from the procedure to qualify as a surgical site infection (Stevens et al., Clin Infect Dis 2014, 59: e10-52).

23:

A temperature of 101.3°F(38.5°C) or greater when associated with a surgical site infection is indicative of a significant systemic response. Hospitalization is required for incision and drainage and intravenous anti-infectives (Stevens et al., Clin Infect Dis 2014, 59: e10-52).

24:

Medical prosthetics or implants are types of medical devices that enhance, support, or replace existing impaired or missing body parts or functions. These devices may be permanent or temporary and composed of man-made or biologic materials. Indwelling (vs transcutaneous) implants pose a greater risk for the development of cellulitis complicated by bacteremia (U.S. Food and Drug Administration, Implants and Prosthetics. 2023).
Some examples of prosthetics or implants are (U.S. Food and Drug Administration, Implants and Prosthetics. 2023):

- breast implants
- knee/hip joint prosthesis
- cardiovascular implantable electronic devices (e.g., permanant pacemaker (PPM))
- surgical mesh repair
- central nervous system shunts

25:

The 2021 Surviving Sepsis Campaign Guidelines recommend timely administration of antimicrobials to patients with septic shock and high likelihood of sepsis (ideally within 1 hour), and to patients with possible sepsis without shock when alternative diagnoses are felt to be less likely after an initial evaluation (within 3 hours) (Evans et al., Crit Care Med 2021, 49: e1063-e143).

26:

Incision and drainage are the most effective treatments for cutaneous abscesses (Schmitz and Gottlieb, Ann Emerg Med 2021, 78: 44-8; Stevens et al., Clin Infect Dis 2014, 59: e10-52).

27:

Examples of vascular imaging may include:

- Computed tomography Angiogram (CTA)
- Magnetic resonance angiogram (MRA)

28:

A noninvasive test to assess for vascular function may include any of the following:

- Ankle-brachial index (ABI)
- Arterial doppler study with pulse volume recordings
- Toe systolic pressure
- Transcutaneous oxygen tension ($TcPO_2$)

29:

Patients with diabetes are at significant risk for developing peripheral arterial disease (PAD). PAD can impede the healing of a diabetic foot ulcer and increase the risk of limb loss. All patients with diabetes who have a foot ulcer should have pedal perfusion evaluation performed (American Diabetes Association, Diabetes Care 2024, 47: S1-S314). Vascular assessment using toe systolic pressure has shown to be more accurate, especially when combined with pulse volume recordings via doppler ultrsound. These patients should have an urgent vascular surgery evaluation. Noninvasive test to assess for pedal perfusion may include any of the following (Hingorani et al., J Vasc Surg 2016, 63: 3S-21S):

- Ankle-brachial index (ABI)
- Arterial doppler study with pulse volume recordings
- Toe systolic pressure
- Transcutaneous oxygen pressure ($TcPO_2$)

30:

Imaging for osteomyelitis may include:

- Plain radiograph
- Magnetic resonance imaging (MRI)
- Bone scan

31:

A plain radiograph is recommended in patients with a diabetic foot ulcer to identify any abnormalities of the bone. Magnetic resonance imaging (MRI) is the imaging of choice when the diagnosis of osteomyelitis remains uncertain, or more sensitive or specific information is necessary. When an MRI is unavailable or contraindicated, a triple-phase bone scan may be performed (Lipsky et al., Diabetes Metab Res Rev 2020, 36 Suppl 1: e3280; Chastain et al., Clin Podiatr Med Surg 2019, 36: 381-95).

32:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.
For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.
NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.
(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

33:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

34:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

35:

In most patients hospitalized with infection, clinical improvement would be expected to occur within 2 days. For patients who are deteriorating or not responding to treatment, it may be necessary to broaden anti-infective coverage while awaiting the results of additional diagnostic tests. In addition, consultation by an infectious disease specialist and further diagnostic tests may be valuable to evaluate reasons for the lack of improvement.

36:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

37:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.
For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.
Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status;** those who do not meet criteria would be appropriate for secondary review.
(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

38:

Although the vast majority of wound care is readily handled in the post-acute setting, there are situations when a patient requires care at the acute level, such as high-frequency wound care required, prolonged length of time required performing the procedure, and the type or amount of medication required to keep the patient comfortable during the procedure. For children less than 12 years of age (or less than 30 kg), the wound care regimen may need to be adjusted to a lesser frequency and/or duration (e.g., twice per day, 10 minutes), though still requiring the current level of care, due to the patient's size, pain tolerance, fluid shifts, and vital sign lability.
To determine the appropriate discharge destination, an assessment is required of the patient and/or caregiver's ability to learn the necessary wound care regimen, complexity of the treatment regimen, and a plan for managing contributory factors. For discharge to the home setting, the patient and/or caregiver must be able and willing to participate in the care plan, or there is sufficient agency assistance available to adequately meet the patient's needs. Patient and/or caregiver instruction should include all the following (Bryant and Nix, Acute & chronic wounds: current management concepts, 5th ed. 2015, xiv, 655 p.):

- Wound care regimen that includes a plan to monitor patient-specific contributory factors
- Adjunctive therapies (medications, nutritional plan, and supplements)
- Re-ordering medications and wound management supplies
- When to seek medical care (e.g., signs, symptoms, and complications to report to the health care provider)
- Continued follow-up and support with health care providers

39:

Pain is considered to be adequately controlled when a stable medication regimen and/or adjunct therapy provides sufficient, symptomatic relief without causing significant adverse side effects. Studies have shown epidural analgesia to be very effective in the initial post-operative period (McNicol et al., Cochrane Database Syst Rev 2015: CD003348; Popping et al., Ann Surg 2014, 259: 1056-67). For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., Jama 2016, 315: 1624-45; Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based, medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended-release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

For patients in the early post-surgical phase, a large retrospective cohort study found that the duration of the opioid prescription, rather than its dosage, was more strongly associated with opioid misuse (Brat et al., BMJ 2018, 360: j5790).

40:

Hyperbaric oxygen therapy (HBO) is used as an adjunct treatment for chronic wound care by increasing cellular oxygen. There is evidence in support of HBO therapy in the treatment of diabetic foot lesions and in the presence of chronic limb-threatening ischemia (CLTI) in diabetic patients (Conte et al., Eur J Vasc Endovasc Surg 2019, 58: S1-S109.e33). Specifically, hyperbaric treatment may be beneficial in preventing amputation and promoting complete healing in patients with Wagner Grade 3 or greater diabetic foot ulcers who have undergone surgical debridement or have shown no significant improvement after 30 or more days of conservative care. For Wagner ulcers Grade 2 or lower, there is insufficient evidence to support HBO treatment. There is also insufficient evidence to support this treatment for arterial wounds and pressure ulcers (Mathieu et al., Diving Hyperb Med 2017, 47: 24-32; Kranke et al., Cochrane Database Syst Rev 2015: CD004123). There is some evidence that HBO therapy improves outcomes in late radiation tissue injury of soft tissue and bones of the head, neck, anus, and rectum (Bennett et al., Cochrane Database Syst Rev 2016, 4: CD005005). The Undersea and Hyperbaric Medical Society (UHMS) recommends treatment with HBO for the following wounds: delayed radiation injury (osteoradionecrosis), necrotizing soft tissue infections, thermal burns, compromised skin grafts and flaps, crush injury, clostridial myositis, myonecrosis (gas gangrene), and refractory osteomyelitis (Undersea Hyperb Med 2018, 45: 379-80). Hyperbaric oxygen may help reduce infection and the need for drainage in crush injuries (Yamada et al., Undersea Hyperb Med 2014, 41: 283-9). Overall, the evidence supporting hyperbaric oxygen for traumatic injuries (including burns) and surgical injuries is limited, and more study is needed (Eskes et al., Cochrane Database Syst Rev 2013, 12: CD008059).

41:

Anti-infective medications may cause adverse drug reactions. Symptoms of an adverse drug reaction include urticaria, angioedema, bronchospasm, pruritus, rash, vomiting, diarrhea, anaphylaxis, and thrombocytopenia.

42:

Guidelines recommend preprandial glucose targets of 80-130 mg/dL (4.4-7.2 mmol/L) and a random blood glucose target range of 100-180 mg/dL (5.6-10.0 mmol/L) in hospitalized patients. In stable patients with previous tight glycemic control, more stringent targets may be appropriate. In patients with severe comorbidities, a history of hypoglycemia, terminal illness, late onset diabetes, and in older patients, less stringent targets may be appropriate. The preferred method for achieving and maintaining glucose control in non-critically ill patients is scheduled subcutaneous insulin administration. Guidelines discourage the sole use of sliding scale insulin in the inpatient hospital setting. Rather, an insulin regimen consisting of basal, prandial, and correctional insulin is recommended in hospitalized patients (American Diabetes Association, Diabetes Care 2024, 47: S1-S314).

43:

Instruction: Insulin adjustments should be based on blood glucose values obtained by lab draw or glucose monitor. Intermittent insulin must be administered at least 3 times in 24 hours. If the patient is on a continuous insulin infusion, the rate must be adjusted at least twice in 24 hours. A dose that is held based on a normal or low blood glucose value should be included in the count of doses administered within the 24-hour period.

44:

Clostridioides difficile is the most common cause of nosocomial infectious diarrhea, particularly in patients receiving anti-infectives (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

45:

Clostridioides difficile is the most common cause of nosocomial infectious diarrhea, particularly in patients receiving anti-infectives. Guidelines describe multiple acceptable regimens but do indicate a preference for oral fidaxomicin for patients with mild to moderate disease. Alternative regimens are driven by disease severity and the number of episodes the patient has experienced. Treatment guidelines recommend discontinuation of the inciting anti-infective as soon as possible to prevent recurrence. Other interventions include IV fluids, electrolytes, or oral fluids to maintain hydration. Use of opiates and antidiarrheal medications should be avoided because decreased intestinal motility will exacerbate the infection (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

46:

Instruction: This criteria point is intended for anti-infectives initiated for the treatment of *C. difficile* infections. The preferred treatment of *Clostridioides difficile* infections is based on the episode of illness and the severity of the symptoms. The Infectious Diseases Society of America (IDSA) recommends a variety of treatment regimens which are dependent on clinical presentation, including disease severity and how many episodes of disease the patient has experienced (Johnson et al., Clin Infect Dis 2021, 73: e1029-e44).

47:

Inadequate oral intake refers to the inability to maintain hydration according to a patient's documented fluid goal or maintenance fluid requirements. The inability to maintain hydration may occur when excessive output (e.g., vomiting, diarrhea, or insensible losses) leads to a net fluid deficit (Scales, Br J Nurs 2014, 23).

48:

Wound care teaching should include instruction on the following:

- Adjunctive therapies (medications, nutritional plan, and supplements)
- Follow-up plan of care
- Medications and wound management supplies
- When to seek medical care (e.g., signs, symptoms, and complications to report to the health care provider)
- Wound care regimen (including patient-specific factors)

49:

Pain assessment includes quality, intensity, location, frequency, duration, and exacerbating or alleviating factors. Uncontrolled pain should be monitored, assessed, and documented. Appropriate medication adjustment and other interventions should be taken to reduce pain.

50:

Instruction: For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

51:

For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

52:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

53:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

54:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

55:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

56:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

57:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk

- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

58:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

59:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

60:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

61:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.
Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

62:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

63:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

64:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

65:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

66:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

67:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g.,

mobility, activities of daily living (ADLs)) may vary.
The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

68:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

69:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

70:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

71:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing

- Impaired respiratory drive
- Inspiratory muscle weakness

72:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.